

Week 4 Report: Regression and Classification Models

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Overview

This week covered two supervised learning tasks: a Linear Regression model predicting house prices from real King County, WA housing sales data, and a Logistic Regression model predicting Titanic passenger survival. Together they cover the two core supervised learning problem types: regression (predicting a continuous number) and classification (predicting a category).

Task 1: House Price Prediction (Linear Regression)

Dataset

4,600 real home sales from King County / Seattle, WA (data.csv), with features including bedrooms, bathrooms, square footage, floors, condition, waterfront and view status, year built, and city.

Data Preparation

- Checked for missing values and duplicates — none found.
- Found and removed 49 rows with a price of \$0 (not real sales), leaving 4,551 rows.
- Ran a correlation matrix across all numeric columns to identify the strongest predictors of price.
- Grouped the 44 unique cities into the 10 most frequent, with the remainder bucketed as "Other," then one-hot encoded the result — keeping the feature space manageable while preserving location signal.
- Selected 9 base features (expanding to 18 after encoding): bedrooms, bathrooms, sqft_living, floors, condition, waterfront, view, yr_built, and city group.

Key Finding: What Drives Price

The correlation analysis showed sqft_living (0.45) as the strongest single predictor, followed by sqft_above (0.38) and bathrooms (0.34). Notably, condition (0.04) and year built (0.02) showed almost no relationship with price in this dataset — a useful, slightly counterintuitive finding worth flagging rather than assuming.

Model and Results

An 80/20 train-test split (3,640 training rows, 911 test rows) was used to train a Linear Regression model.

Metric	Value
RMSE	\$227,589.82
MAE	\$140,623.76
R ²	0.652

The model explains about 65% of the variation in house prices. The gap between RMSE and MAE points to a small number of large errors — driven by a handful of high-value outlier homes — pulling the average error up. This is an honest, expected result for plain linear regression on real, noisy housing data, and a natural lead-in to trying non-linear models (e.g. Random Forest) in future work.

Task 2: Titanic Survival Prediction (Logistic Regression)

Dataset

891 Titanic passenger records, with features including passenger class, sex, age, fare, number of siblings/spouses and parents/children aboard, and port of embarkation.

Data Preparation

- Dropped the Cabin column — over 70% of its values were missing.
- Filled missing Age values with the median, and missing Embarked values with the mode.
- Detected outliers in Fare using the IQR method (116 rows flagged) and capped extreme values rather than deleting them, to preserve sample size.
- Label-encoded Sex (female = 0, male = 1) and one-hot encoded Embarked.
- Applied StandardScaler to Age and Fare so both features contribute on a comparable scale.

Key Finding: What Drives Survival

Correlation with survival showed Sex (-0.54) and Passenger Class (-0.34) as by far the strongest predictors, followed by Fare (+0.32). This lines up with the historical record of the disaster — women and first-class passengers had substantially better survival odds.

Model and Results

An 80/20 train-test split (712 training rows, 179 test rows) was used to train a Logistic Regression model.

Metric	Value
Accuracy	85.5%
Precision (Survived)	0.80
Recall (Survived)	0.77
Precision (Died)	0.88
Recall (Died)	0.90

With roughly 62% of passengers having died, a model that always predicted "died" would score about 62% accuracy without learning anything. At 85.5%, the model is clearly picking up real, meaningful patterns — most notably the strong influence of sex and class on survival odds.

Summary

- Regression (house prices): $R^2 = 0.652$, RMSE = \$227,590. Size and bathroom count are the strongest price drivers; condition and age barely matter in this dataset.
- Classification (Titanic survival): Accuracy = 85.5%. Sex and passenger class are the dominant survival predictors, consistent with historical accounts.

Both models confirm a core ML lesson: feature relevance is empirical, not assumed. Correlation and coefficient analysis in both tasks surfaced which inputs actually mattered, sometimes contradicting intuition (e.g. house condition and age having negligible price impact here).